

Program 1 output:-

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Tasks Console X Eclipse IDE for C/C++ Developers 2026-09 M3
<terminated> (exit value: 0) Assigment1.exe [C/C++ Application] C:\Users\Administrator\
Enter Temp in Degree C :47.3
Status: CRITICAL
Action: Cooling system triggered
Reading: Above Average
Temperature in Fahrenheit: 117.14 °F
<

```

Program 2 output:-

```

Tasks Console X Eclipse IDE for C/C++ Developers 2026-09 M3
<terminated> (exit value: 0) Assigment1.exe [C/C++ Application] C:\Users\Administrator\
Readings entered : 8
Enter 8 readings: 22.1 31.5 -1 46.0 28.0 50.2 10.0 38.0
Valid readings : 22.1 31.5 46 28 50.2 10 38
Skipped (errors) : 1
First CRITICAL : Index 3 -> 46 C
Min : 10 C
Max : 50.2 C
Avg : 32.2571 C
Normal : 3
Warning : 2
Critical : 2
Shutdown : 0

```

Program 3 output:-

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<terminated> (exit value: 0) Assigment1.exe [C/C++ Application] C:\U
temp of 9 rooms:
24.0 31.5 28.0 45.0 22.0 30.0 19.0 27.5 50.2
      Room1   Room2   Room3
Floor 1 : 24   31.5   28
Floor 2 : 45   22     30
Floor 3 : 19   27.5   50.2
Hottest Room : Floor 3, Room 3 -> 50.2 °C
Hottest Floor : Floor 2 (avg 32.3333 °C)
Rooms at WARNING or above : 4

```

Program 4 output:-

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Console X Eclipse IDE for C/C++ Developers 2026-09 M3 Sponsor the Eclipse IDE
<terminated> (exit value: 0) Assigment1.exe [C/C++ Application] C:\Users\Administrator\eclipse-work
Config : Warn=30 °C   Critical=45 °C   Readings=10
Results : Normal:2   Warning:5   Critical:2   Shutdown:1

```

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Console × Eclipse IDE for C/C++ Developers 2026-09 M3 Sponsor the Eclipse IDE
<terminated> (exit value: 1) Assigment1.exe [C/C++ Application] C:\Users\Administrator\eclipse-workspace\Assigment1\
Usage : ./sensor_monitor <warn_threshold> <critical_threshold> <num_readings>
Error : Missing arguments.

```

Program 5 output:-

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<terminated> (exit value: 0) Assigment1.exe [C/C++ Application] C:\Users\Administrator\
--- V1: Call by Value ---
Before : A=55 B=12
After  : A=55 B=12    <- values unchanged

--- V2: Call by Reference ---
Before : A=55 B=12
After  : A=12 B=55    <- values swapped

--- V3: Call by Pointer ---
Before : A=12 B=55
After  : A=55 B=12    <- values swapped back

```

Program 6 output:-

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Console × Eclipse IDE for C/C++ Developers 2026-09 M3 Sponsor the Eclipse IDE
<terminated> (exit value: 0) Assigment1.exe [C/C++ Application] C:\Users\Administrator\eclipse-workspace\
Original Signal : 0.5 -1.2 0.8 -0.3 1 -0.9 0.1
RMS : 0.778276
Zero Crossings : 6
After Normalise : 0.416667 -1 0.666667 -0.25 0.833333 -0.75 0.0833333
After Gain : 0.833333 -2 1.33333 -0.5 1.66667 -1.5 0.166667

```

Program 7 output:-

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Console × Eclipse IDE for C/C++ Developers 2026-09 M3 Sponsor the Eclipse IDE
<terminated> (exit value: 0) Assigment1.exe [C/C++ Application] C:\Users\Administrator\eclipse-
Status Register : 177
Control Register Before : 0
Control Register After : 25
Status Register : 177

```

Program 8 output:-

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Console × Eclipse IDE for C/C++ Developers 2026-09 M3 Sponsor the Eclipse IDE
<terminated> (exit value: 0) Assigment1.exe [C/C++ Application] C:\Users\Administrator\
Calibration Min : 8
Calibration Max : 67

```

Program 9 output:-

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Console × Eclipse IDE for C/C++ Developers 2026-09 M3 Sponsor the B
<terminated> (exit value: 0) Assigment1.exe [C/C++ Application] C:\Users\Administrato
Waypoint 1 : Distance = 50
Inside Safe Zone : Yes
Waypoint 2 : Distance = 63.2456
Inside Safe Zone : No
Waypoint 3 : Distance = 22.3607
Inside Safe Zone : Yes
90 degrees in radians : 1.5708
Clamp 75 between 0 and 50 : 50

```

Program 10 output:-

```

Console × Eclipse IDE for C/C++ Developers 2026-09 M3 Sponsor the B
<terminated> (exit value: 0) Assigment1.exe [C/C++ Application] C:\Users\Administrato
Enter name: Tanaya
Enter department: HR
Enter grade: B
Enter basic salary: 75000
Enter name: Apurva
Enter department: Finance
Enter grade: C
Enter basic salary: 90000
Enter name: Gaurav
Enter department: Engineering
Enter grade: A
Enter basic salary: 60000
=====
EMPLOYEE PAYSALIP - AUG 2026
=====
Emp ID      : 1001
Name        : Tanaya
Department  : HR
Grade       : B
Status      : Active
-----
Basic Salary      : Rs. 75000.00
Allowances (30%) : Rs. 22500.00
Gross Salary     : Rs. 97500.00
-----
Tax Deduction    : Rs. 4750.00
Net Salary       : Rs. 92750.00
=====
=====

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Console × Eclipse IDE for C/C++ Developers 2026-09 M3 Sponsor the Ed
<terminated> (exit value: 0) Assignment1.exe [C/C++ Application] C:\Users\Administrator\

=====
EMPLOYEE PAYSIP - AUG 2026
=====
Emp ID      : 1002
Name        : Apurva
Department  : Finance
Grade       : C
Status      : Active
-----
Basic Salary : Rs. 90000.00
Allowances (20%) : Rs. 18000.00
Gross Salary : Rs. 108000.00
-----
Tax Deduction : Rs. 6600.00
Net Salary    : Rs. 101400.00
=====
EMPLOYEE PAYSIP - AUG 2026
=====
Emp ID      : 1003
Name        : Gaurav
Department  : Engineering
Grade       : A
Status      : Active
-----
Basic Salary : Rs. 60000.00
Allowances (40%) : Rs. 24000.00
Gross Salary : Rs. 84000.00
-----
Tax Deduction : Rs. 3400.00
Net Salary    : Rs. 80600.00
=====

```

```

=====
Gaurav is no longer active. Payroll skipped.
Total Employees : 3

```